

IIIT BHUBANESWAR

Theory Of Computation
(Assignment Solutions)

BTech (CSE – A)

May - 2026

Instructor: Dr. Ajaya Kumar Dash



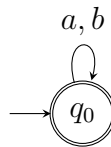
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Assignment Problems

Ques 1

Represent the regular expression that derives all the strings over $\{a, b\}$ including ϵ .

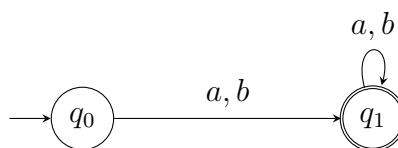
Solution:

$$(a \cdot b)^*$$


Ques 2

Represent the regular expression that derives all the strings over $\{a, b\}$ excluding ϵ .

Solution:

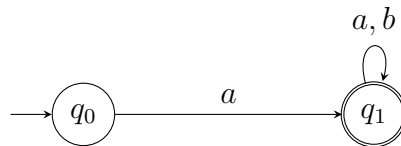
$$(a + b)(a + b)^*$$


Ques 3

Represent the regular expression that derives all the strings over $\{a, b\}$ where every string starts with a .

Solution:

$$a(a + b)^*$$

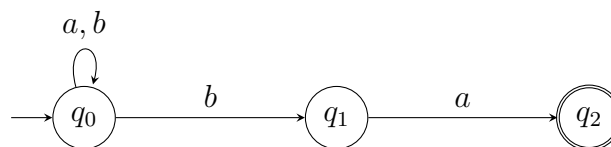


Ques 4

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string ends with “ba”.

Solution:

$$(a + b)^*ba$$

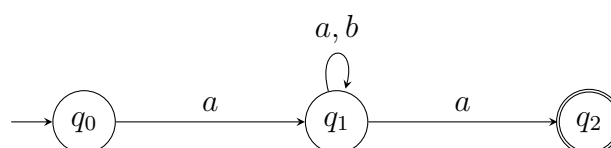


Ques 5

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string starts and ends with a .

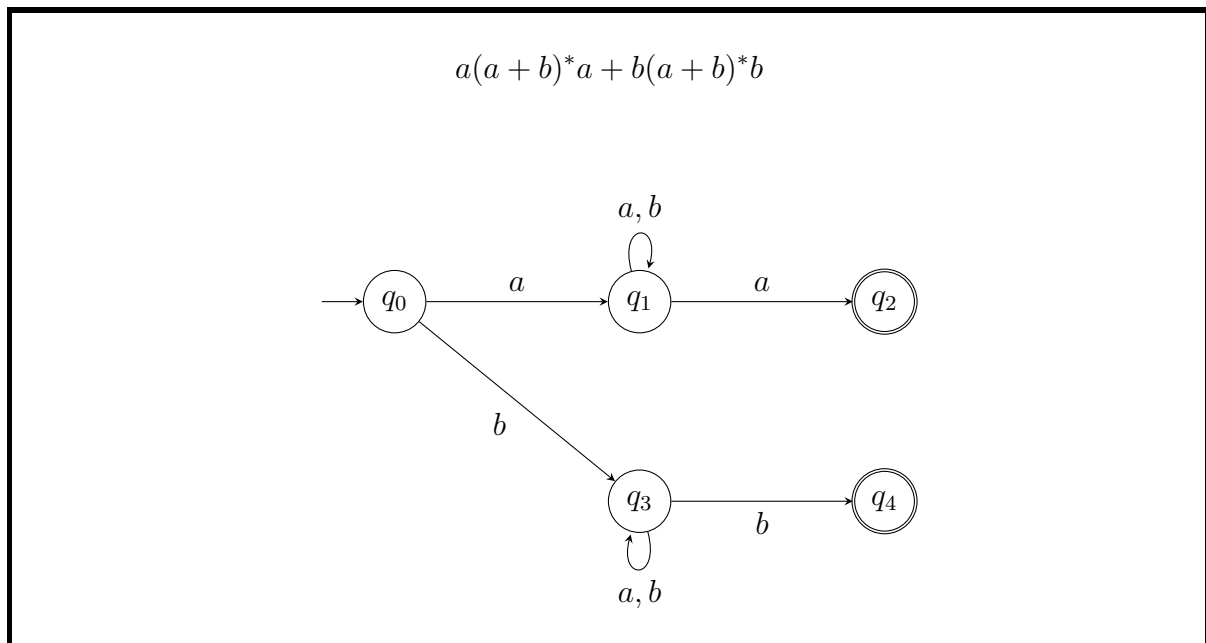
Solution:

$$a(a + b)^*a$$



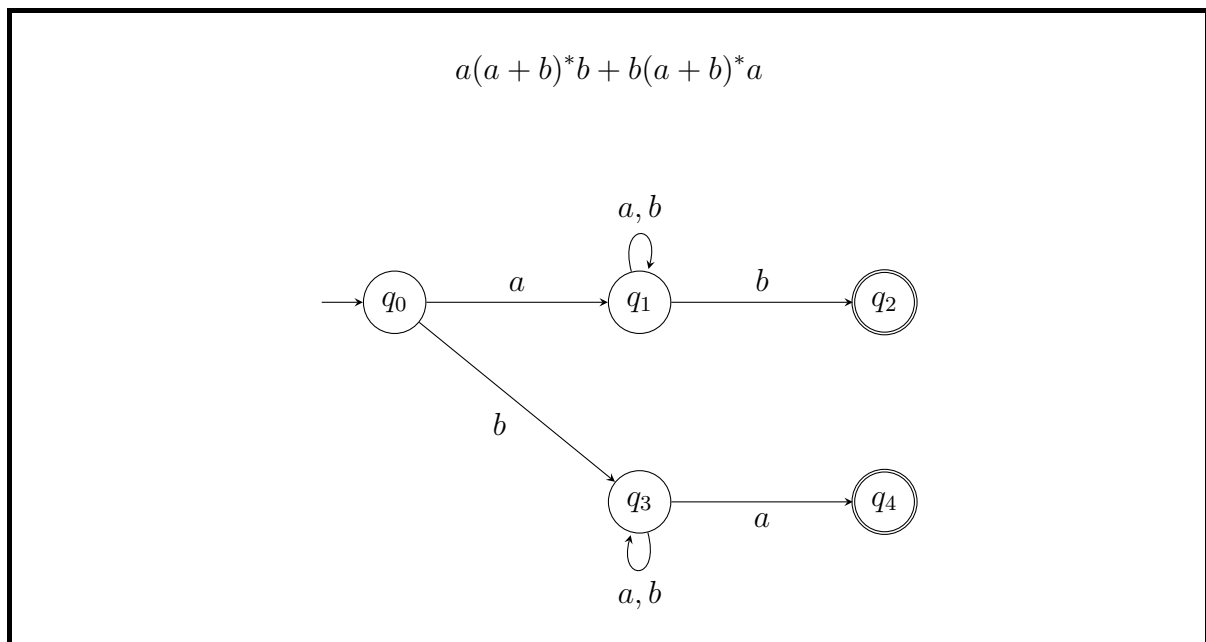
Ques 6

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string starts and ends with the same symbol.



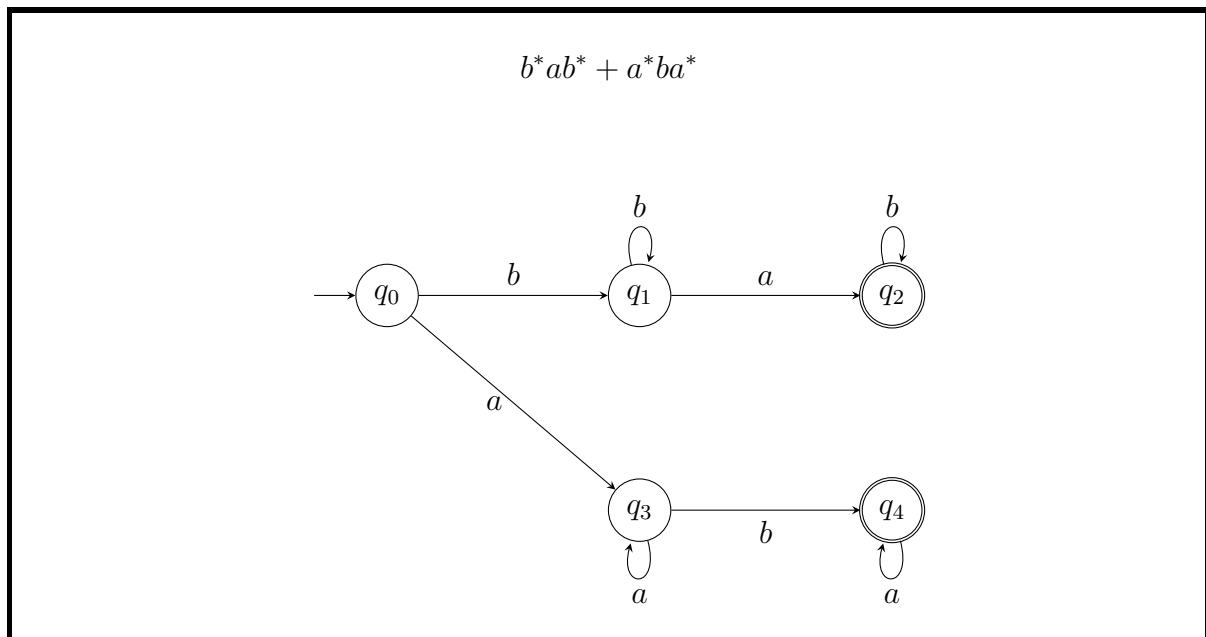
Ques 7

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string starts and ends with different symbols.



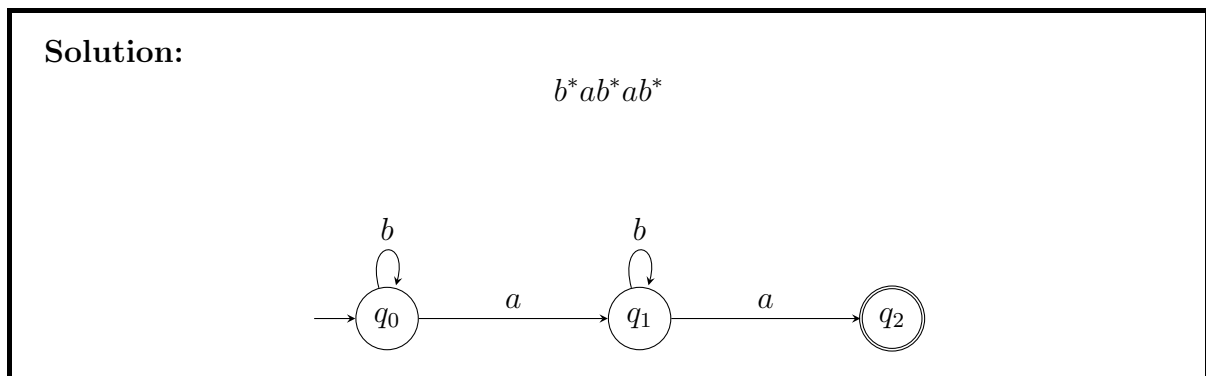
Ques 8

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string contains exactly one a or exactly one b .



Ques 9

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string contains exactly two a 's.

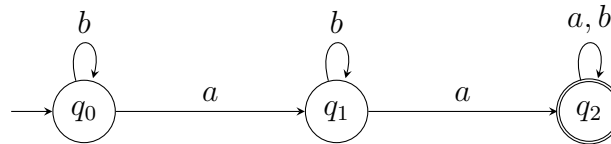


Ques 10

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string contains at least two a 's.

Solution:

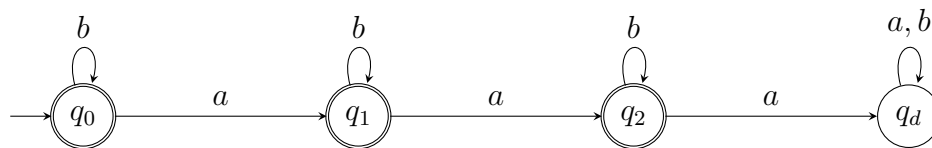
$$b^*ab^*a(a+b)^*$$



Ques 11

Represent the regular expression that derives all strings over $\{a, b\}$ where each string contains at most two a 's.

$$b^* + b^*ab^* + b^*ab^*ab^*$$

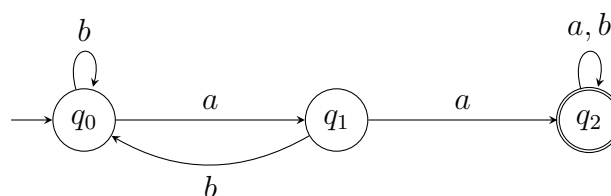


Ques 12

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string contains two consecutive a 's.

Solution:

$$(a+b)^*aa(a+b)^*$$

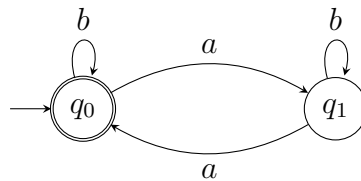


Ques 13

Represent the regular expression that derives the set of all strings over $\{a, b\}$ where each string contains an even number of a 's.

Solution:

$$(b^*ab^*ab^*)^*$$

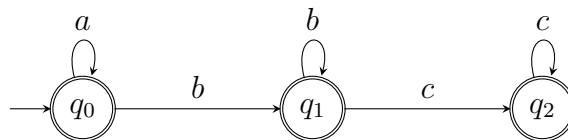


Ques 14

Represent the regular expression that derives the set of all strings over $\{a, b, c\}$ where each string contains any number of a 's followed by any number of b 's followed by any number of c 's.

Solution:

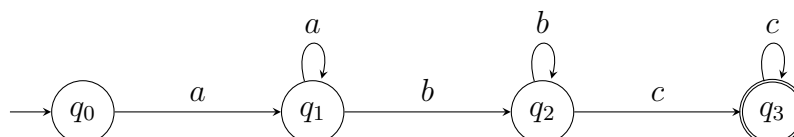
$$a^*b^*c^*$$



Ques 15

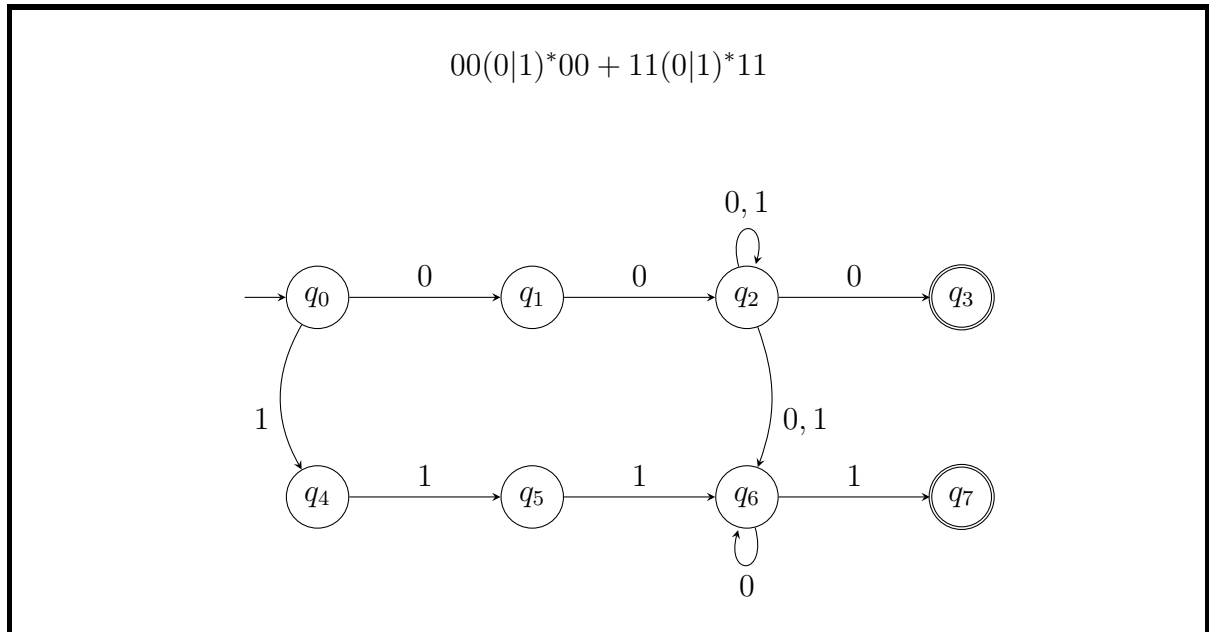
Represent the regular expression that derives the set of all strings over $\{a, b, c\}$ where each string contains at least one a , followed by at least one b , followed by at least one c .

$$a^+b^+c^+$$



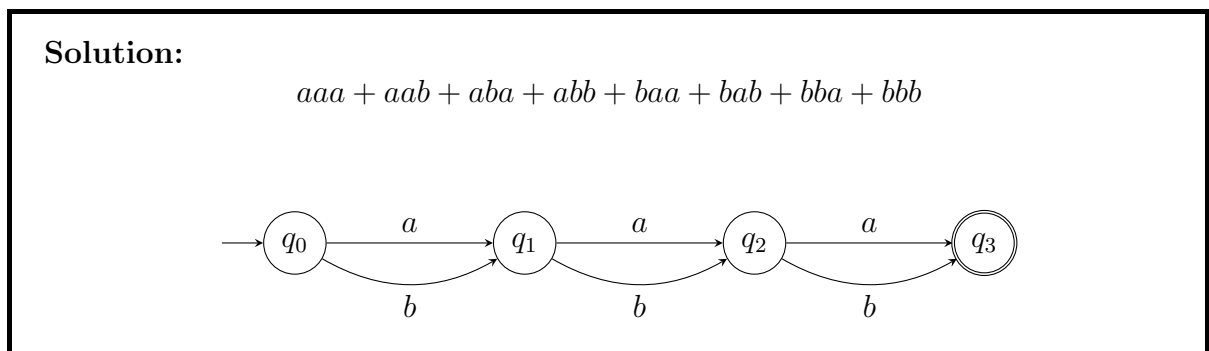
Ques 16

Represent the regular expression that derives the set of all strings over $\{0, 1\}$ where each string begins and ends with the same fixed block (either “00” or “11”).



Ques 17

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is exactly 3.

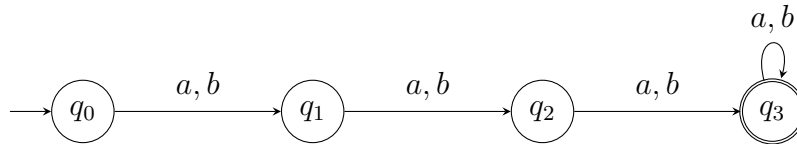


Ques 18

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is at least 3.

Solution:

$$(a + b)(a + b)(a + b)(a + b)^*$$

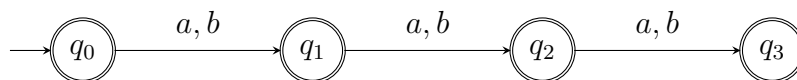


Ques 19

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is at most 3.

Solution:

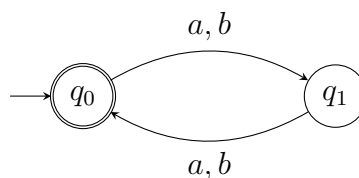
$$\epsilon + (a + b) + (a + b)(a + b) + (a + b)(a + b)(a + b)$$



Ques 20

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is even.

$$((a + b)(a + b))^*$$

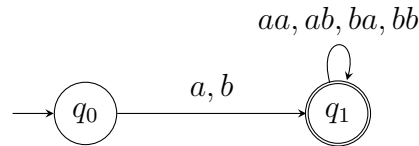


Ques 21

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is odd.

Solution:

$$(a|b)((a|b)(a|b))^*$$

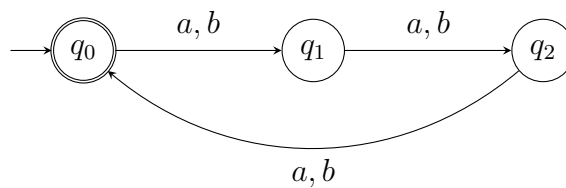


Ques 22

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is divisible by 3.

Solution:

$$((a|b)(a|b)(a|b))^*$$

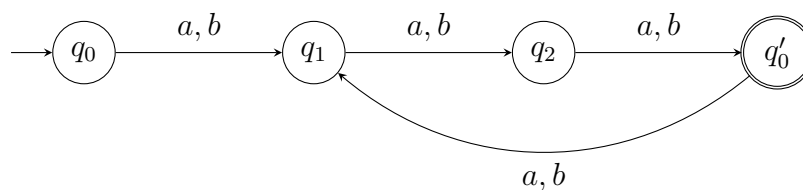


Ques 23

Represent the regular expression that derives all strings over $\{a, b\}$ whose length is divisible by 3 (excluding ϵ).

Solution:

$$((a|b)(a|b)(a|b))^+$$

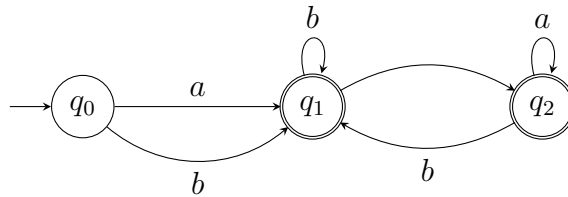


Ques 24

Represent the regular expression that derives all strings over $\{a, b\}$ that do not contain two consecutive identical symbols.

Solution:

$$a(ba)^* + b(ab)^*$$

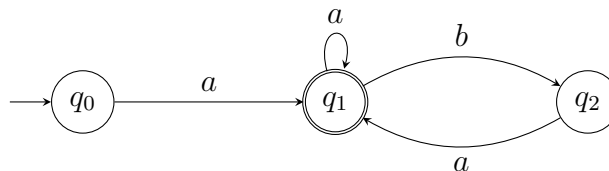


Ques 25

Represent the regular expression that derives all strings over $\{a, b\}$ that start with a and do not have two consecutive b 's.

Solution:

$$a(a + ab)^*$$

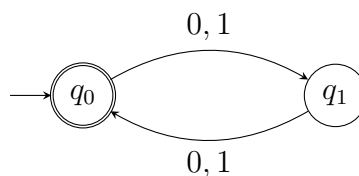


Ques 26

Represent the regular expression that derives all strings over $\{0, 1\}$ with even length.

Solution:

$$(00 + 01 + 10 + 11)^*$$

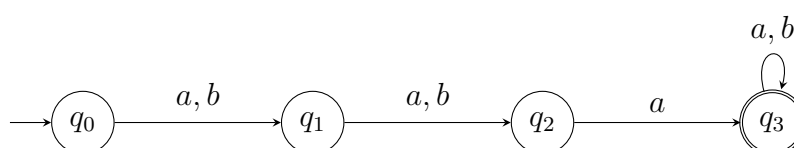


Ques 27

Represent the regular expression that derives all strings over $\{a, b\}$ where the third character from the right is always a .

Solution:

$$(\Sigma - \{a\})^* a \Sigma \Sigma \quad \text{with } \Sigma = \{a, b\}$$

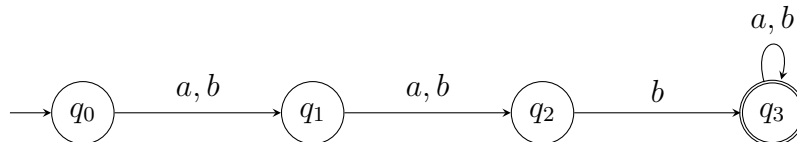


Ques 28

Represent the regular expression that derives all strings over $\{a, b\}$ where the third character from the left is always b .

Solution:

$$\Sigma\Sigma b\Sigma^* \quad \text{with } \Sigma = \{a, b\}$$



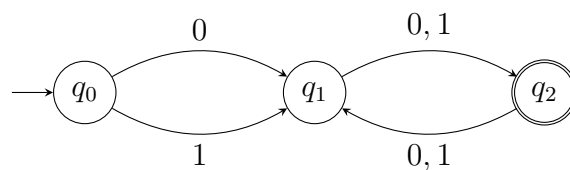
Ques 29

Represent the regular expression that derives all strings over $\{0, 1\}$ having the following two characteristics:

1. Strings that start with 0 and have odd length.
2. Strings that start with 1 and have even length.

Solution:

$$0(00 + 01 + 10 + 11)^* + 1(0 + 1)(00 + 01 + 10 + 11)^*$$

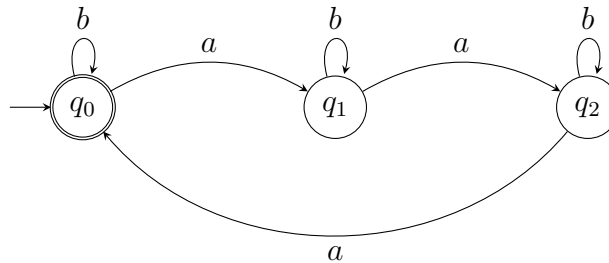


Ques 30

Represent the regular expression that derives all strings over $\{a, b\}$ in which the number of a 's is divisible by 3.

Solution:

$$(b + aa + aba + baa + abba + bab + bba)^*$$

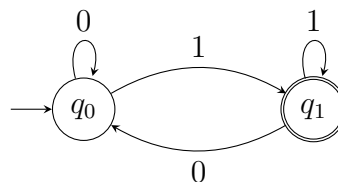


Ques 31

Represent the regular expression for the set of all binary strings (over $\{0, 1\}$) that, when viewed as binary numbers, are not divisible by 2.

Explanation: A binary number is odd if and only if it ending with 1.

$$(0 + 1)^*1$$

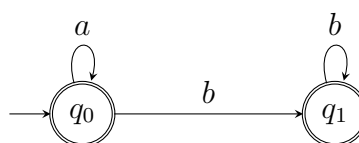


Ques 32

Represent the regular expression for the language $L = \{a^m b^n \mid m, n \geq 0\}$.

Solution:

$$a^*b^*$$

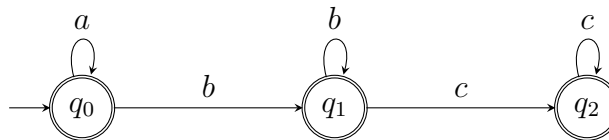


Ques 33

Represent the regular expression for the language $L = \{a^m b^n c^p \mid m, n, p \geq 0\}$.

Solution:

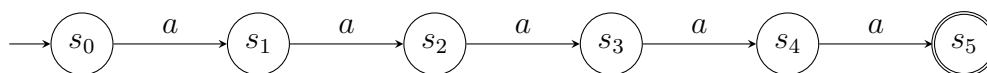
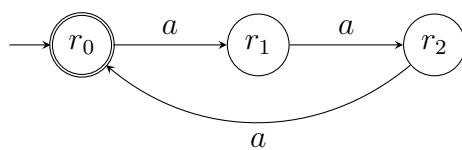
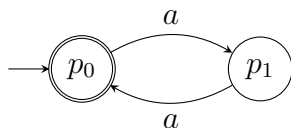
$$a^* b^* c^*$$



Ques 34

Represent the regular expression for the language $L = \{a^n \mid n \text{ is divisible by 3, or } n \text{ is divisible by 5}\}$.

$$(aa)^* + (aaa)^* + aaaaa$$

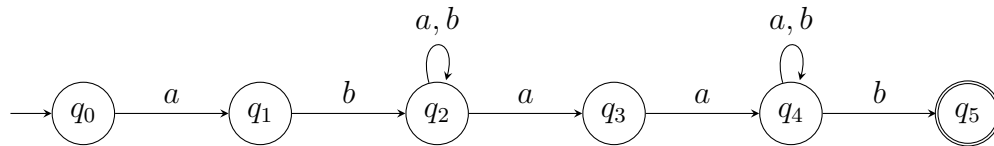


Ques 35

Represent the regular expression that derives all strings over $\{a, b\}$ that start with “ab”, end with “ba”, and contain the substring “aa” somewhere in between.

Solution:

$$ab(a+b)^*aa(a+b)^*ba$$

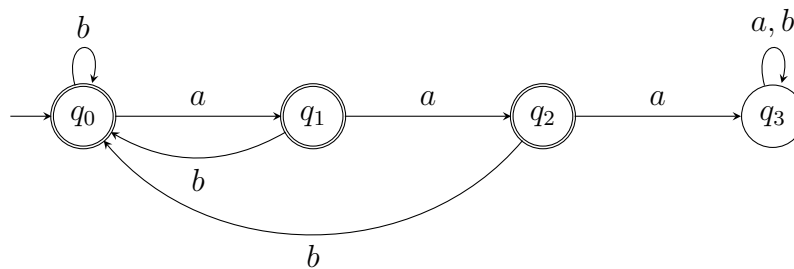


Ques 36

Represent the regular expression for the set of all strings over $\{a, b\}$ that do not end with the substring “aaa”.

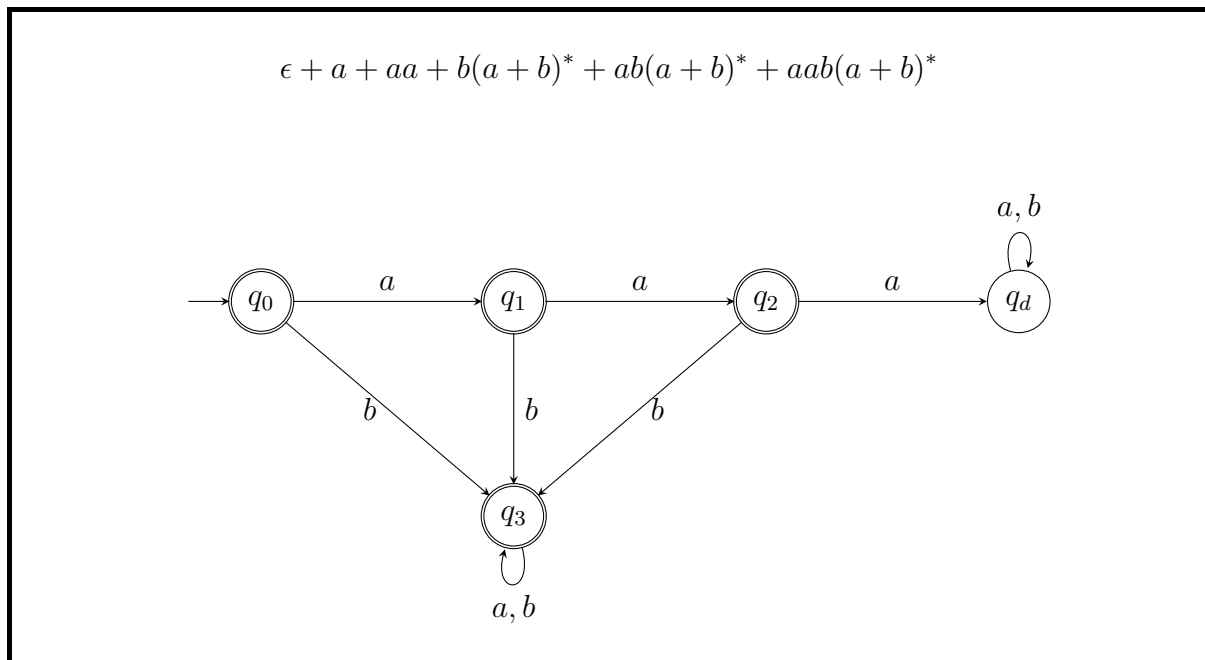
Solution:

$$(a+b)^*(b+ab+aab)$$



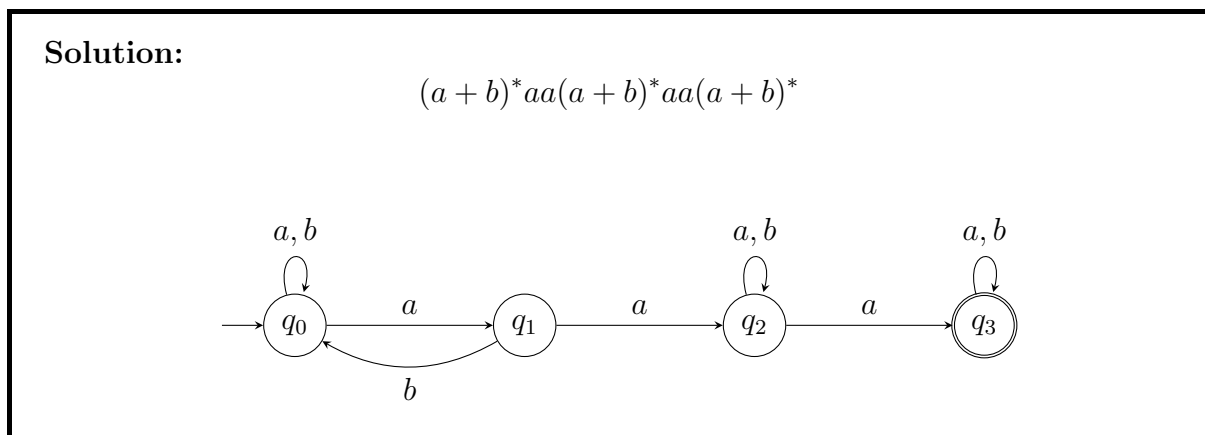
Ques 37

Represent the regular expression for the set of all strings over $\{a, b\}$ that do not begin with three consecutive a 's (i.e., do not start with “aaa”).



Ques 38

Represent the regular expression for the set of all strings over $\{a, b\}$ that contain the substring “aa” at least twice.

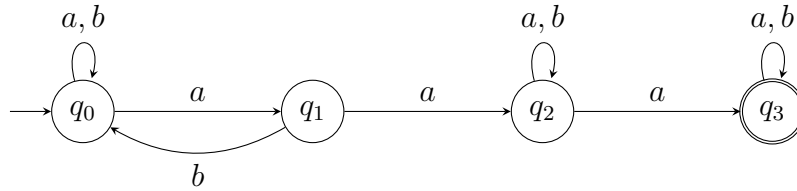


Ques 39

Represent the regular expression for the set of all strings over $\{a, b\}$ that contain the substring “aa” at least twice (alternate formulation).

Solution:

$$(a + b)^* aa(a + b)^* aa(a + b)^*$$

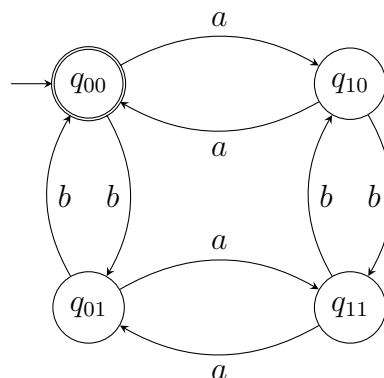


Ques 40

Represent the regular expression for the set of all strings over $\{a, b\}$ in which the total number of a 's is even and the total number of b 's is even.

Solution:

$$(aa + bb + (ab + ba)(ab + ba))^*$$

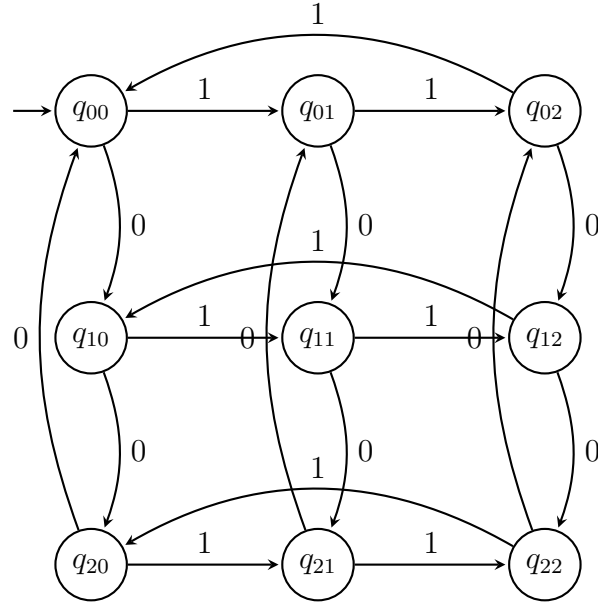


Homework Questions

Ques - 1 :

Construct a minimal DFA over 0's and 1's such that a. $na(w) = 0 \text{ MOD } 3$ and $nb(w) = 1 \text{ mod } 3$ b. $na(w) \text{ mod } 3 = nb(w) \text{ mod } 2$ c. $na(w) \text{ mod } 3 \neq nb(w) \text{ mod } 2$

Solution : We track the count of 0's modulo 3 and the count of 1's modulo 3. This requires a product construction with states (i, j) where i represents $n_0(w) \bmod 3$ and j represents $n_1(w) \bmod 3$.

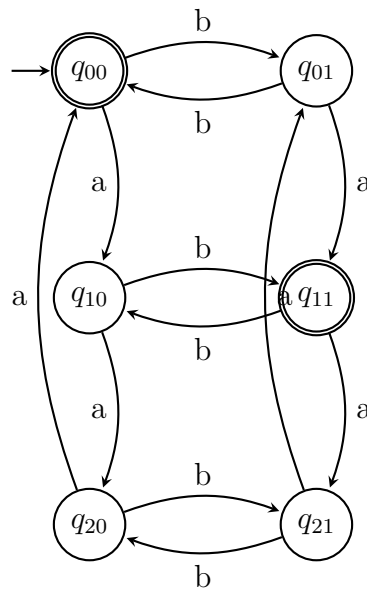


The accepting state is q_{01} (shown with double circle, should be marked) where the first index is 0 (number of 0's $\equiv 0 \pmod{3}$) and second index is 1 (number of 1's $\equiv 1 \pmod{3}$).

Ques - 2 :

Construct a minimal DFA such that $n_a(w) \bmod 3 = n_b(w) \bmod 2$.

Solution: States here tracks $(n_a \bmod 3, n_b \bmod 2)$ and we accepted when both values are equal.



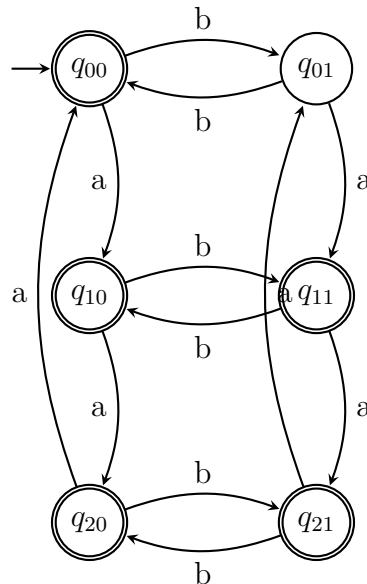
Accept states are q_{00} ($0=0$) and q_{11} ($1=1$).

Ques - 3 :

Construct a minimal DFA such that $n_a(w) \bmod 3 \geq n_b(w) \bmod 2$.

Solution:

We here accept this when $n_a \bmod 3 \geq n_b \bmod 2$.

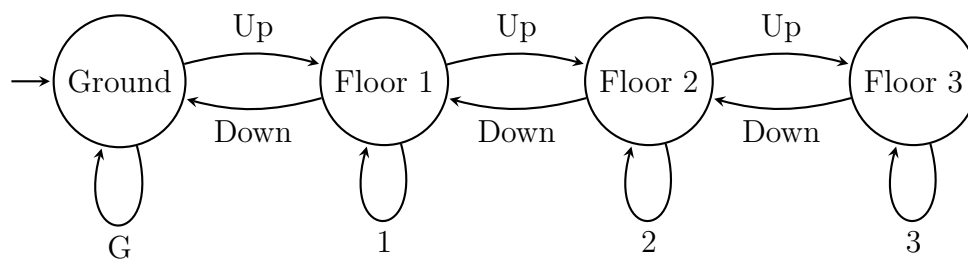


Accept states are $q_{00}, q_{10}, q_{11}, q_{20}, q_{21}$.

Ques - 4 :

Consider the institute building lift, design a state machine for the lift operation in academic building and draw an DFA.

Solution:

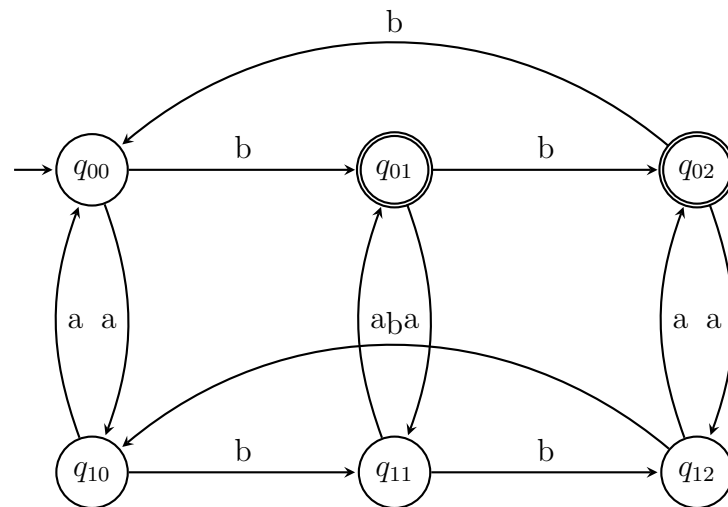


Each state here representing the current floor. Transitions here occurs on Up/Down buttons or by selecting the floor number.

Ques - 5 :

Design a DFA of a's and b's such that number of a's is divisible by 2 and number of b's is not divisible by 3.

Solution:

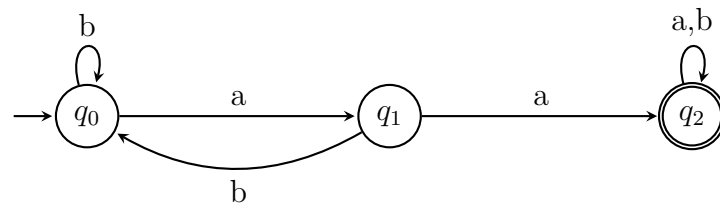


Answer: Accept states: q_{01} and q_{02} (even number of a 's, b 's not divisible by 3).

Ques - 6a:

Construct a DFA of a 's and b 's for the set of all strings containing aa .

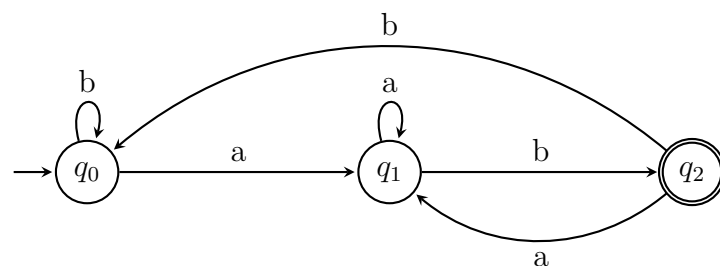
Solution:



Ques - 6b :

Construct a DFA of a 's and b 's for the set of all strings ending with ab .

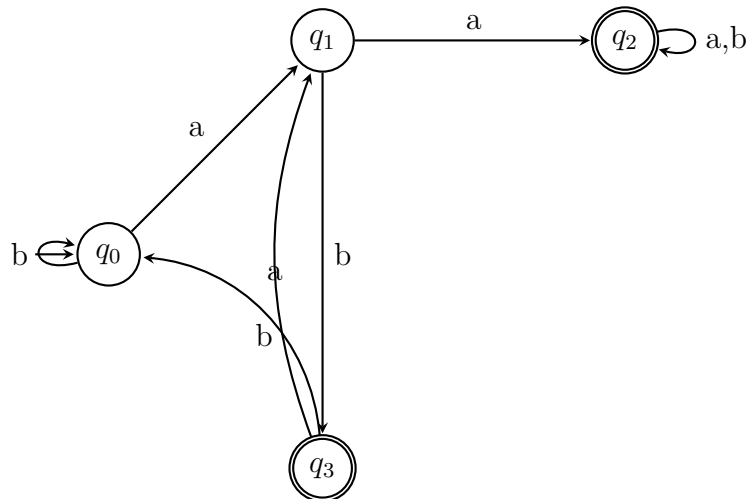
Solution:



Ques - 6c :

Construct a DFA of a's and b's for strings containing aa OR ending with ab .

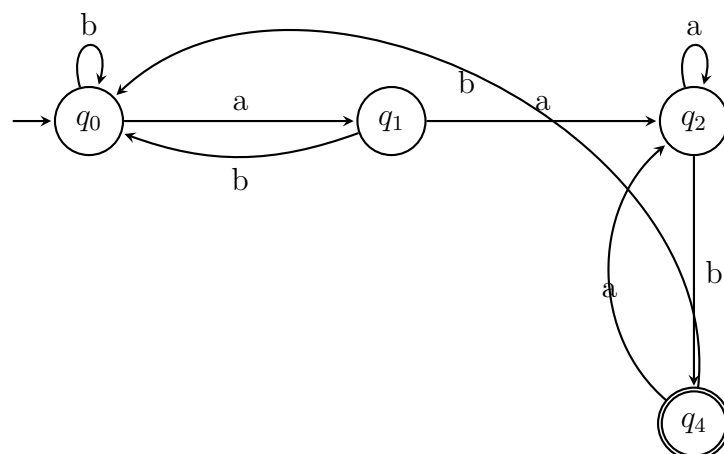
Solution:



Ques - 6d :

Construct a DFA of a's and b's for strings containing aa AND ending with ab .

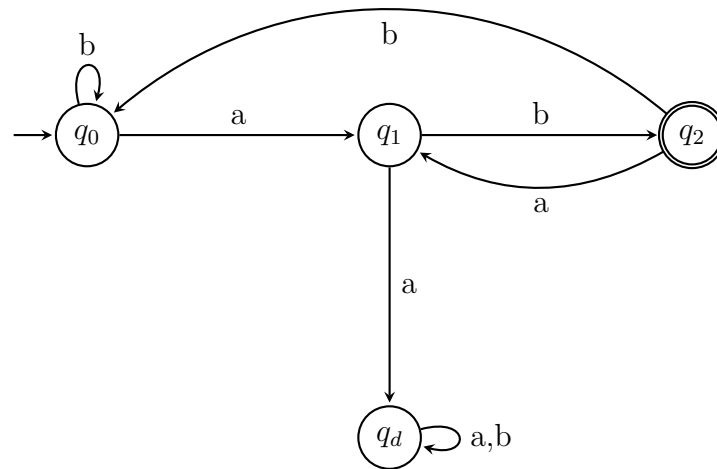
Solution:



Ques - 6e :

Construct a DFA of a's and b's for strings NOT containing aa but ending with ab .

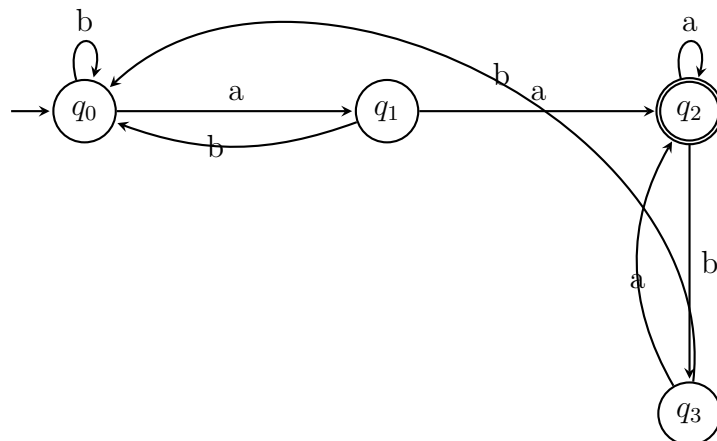
Solution:



Ques - 6f :

Construct a DFA over of a's and b's for strings containing aa but NOT ending with ab .

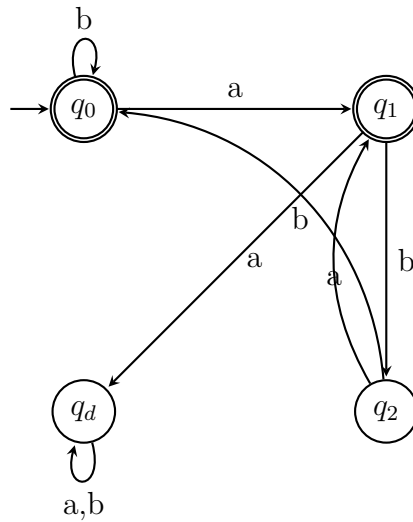
Solution:



Ques - 6g :

Construct a DFA of a's and b's for strings neither containing aa nor ending with ab .

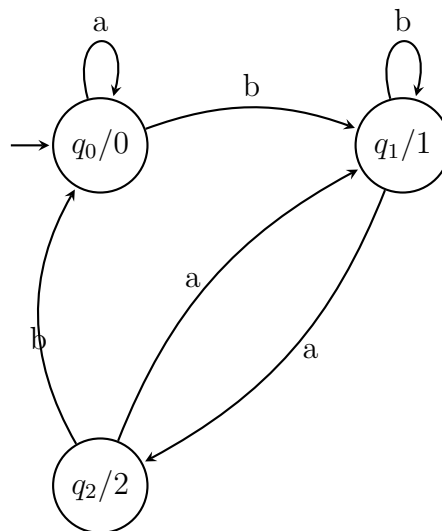
Solution:



Ques - 7 :

Design a Moore machine over alphabet of a's and b's such that the decimal equivalent of binary string is residue modulo 3. (Interpret a=0, b=1)

Solution:



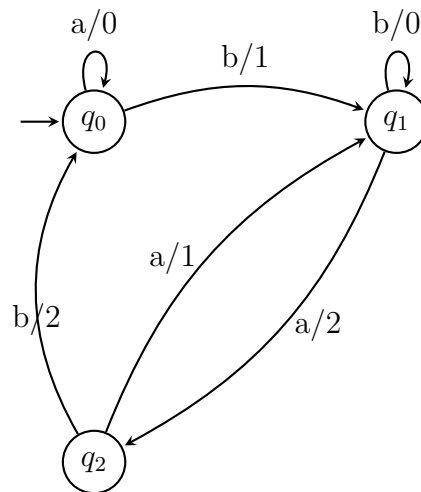
State Output : $q_0 \rightarrow 0, q_1 \rightarrow 1, q_2 \rightarrow 2$

Transition Logic : From state q_v on input x : go to $q_{(2v+x) \bmod 3}$

Ques - 8 :

Design a Mealy machine for Ques 7.

Solution:

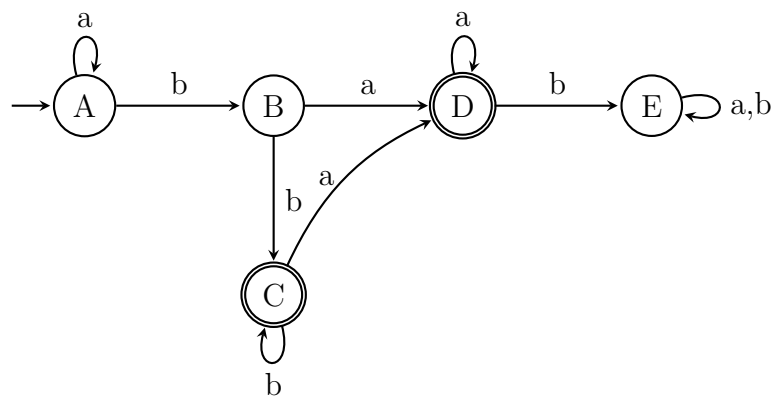


Transition Output : From state q_v on input x : output $(2v + x) \bmod 3$

Ques - 9 :

Write regular expression for the DFA shown in Figure .

DFA from Figure:



Solution:

$$r = a^*b(bb^*a \mid a)a^*$$

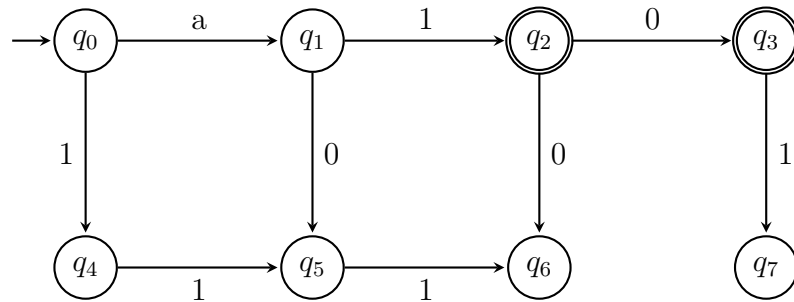
Simplified:

$$r = a^*bb^*aa^*$$

Ques - 10 :

Write regular expression for the NFA shown in Figure .

NFA from Figure:



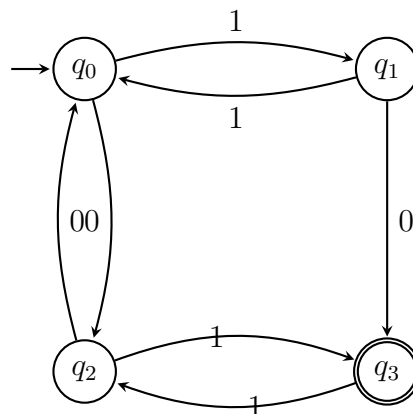
Solution:

$$r = a10 \mid 11(0 \mid 1)$$

Ques - 11 :

Regular expression for the DFA shown in Figure .

DFA from Figure:



Solution:

$$r = (11 \mid 00)^*(10 \mid 01)(11 \mid 00 \mid 10 \mid 01)^*$$

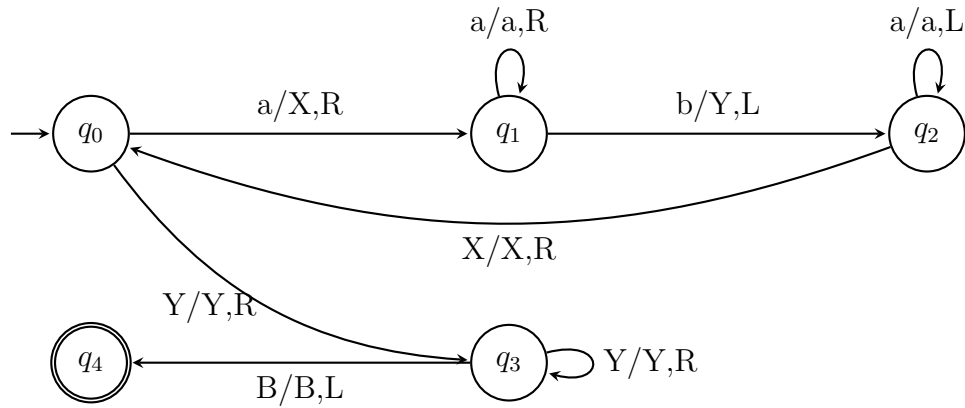
Ques - 12 :

Design a Turing machine which can accept $\{a^n b^n \mid n \geq 1\}$.

Solution:

Algorithm:

1. Mark 1st unmarked a with X
2. Scanning the right to find the first unmarked b and mark it with Y
3. Return to the leftmost position
4. Repeat until all symbols are marked.
5. Accept if pattern is correct is coming.



Transition Function:

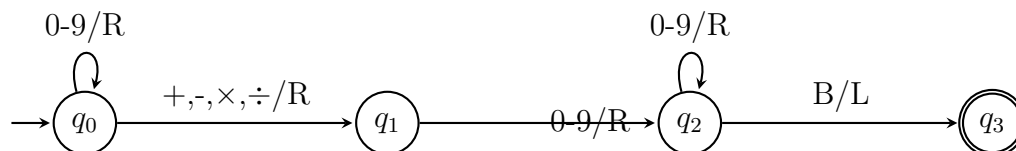
$$\begin{aligned}
 \delta(q_0, a) &= (q_1, X, R) \\
 \delta(q_1, a) &= (q_1, a, R) \\
 \delta(q_1, Y) &= (q_1, Y, R) \\
 \delta(q_1, b) &= (q_2, Y, L) \\
 \delta(q_2, a) &= (q_2, a, L) \\
 \delta(q_2, Y) &= (q_2, Y, L) \\
 \delta(q_2, X) &= (q_0, X, R) \\
 \delta(q_0, Y) &= (q_3, Y, R) \\
 \delta(q_3, Y) &= (q_3, Y, R) \\
 \delta(q_3, B) &= (q_4, B, L)
 \end{aligned}$$

Ques - 13 :

Design a Turing machine that can accept arithmetic operations $\{+, -, \times, \div\}$.

Solution:

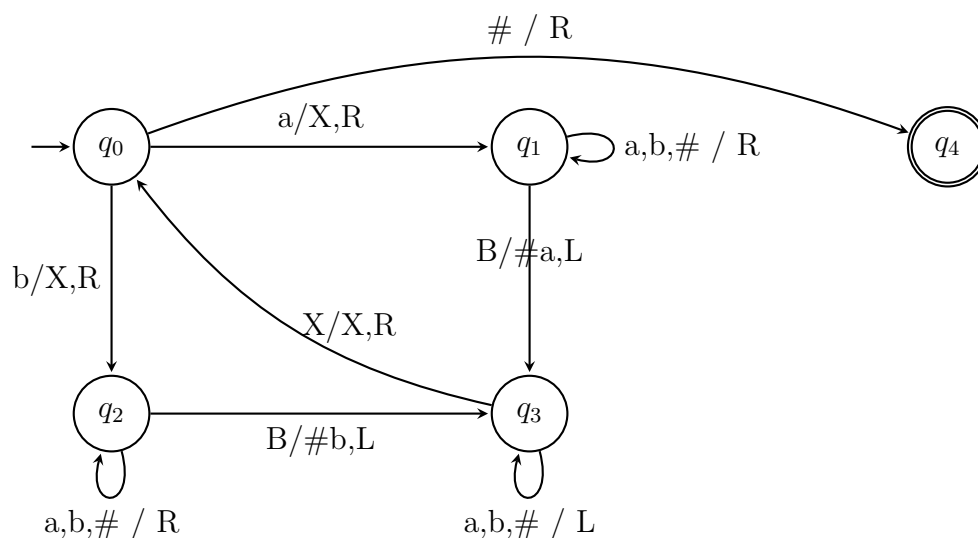
This Turing Machine recognize expressions: $\text{num}_1 \text{ op num}_2$



Ques - 14 :

Copy the content of input - design a Turing machine that produces $w\#w$.

Solution:



Ques - 15a :

Prove $L = \{w \mid n_a(w) < n_b(w)\}$ is not regular where $w \in \Sigma^*$, $\Sigma = \{a, b\}$.

Proof using Pumping Lemma:

Assume L is regular. Let p be the pumping length.

Consider $s = a^p b^{p+1} \in L$ (since $p < p+1$).

By pumping lemma, $s = xyz$ where $|xy| \leq p$, $|y| > 0$, and $xy^i z \in L$ for all $i \geq 0$.

Since $|xy| \leq p$, both x and y consist only of a 's. Let $|y| = t > 0$.

Consider $xy^2 z = a^{p+t} b^{p+1}$.

For this to be in L : $p+t < p+1 \Rightarrow t < 1$.

But $t \geq 1$ (since $|y| > 0$), contradiction

Therefore, L is not regular.

Ques - 15b :

Prove $L = \{ww^R \mid w \in \Sigma^*\}$ is not regular where $\Sigma = \{a, b\}$.

Proof:

Assume L is regular with pumping length p .

Consider $s = a^p b b a^p \in L$.

By pumping lemma, $s = xyz$ where $|xy| \leq p$, $|y| > 0$.

Let $y = a^t$ where $t > 0$.

Then $xy^2z = a^{p+t} b b a^p$.

For $xy^2z = ww^R$, we need first half = reverse of second half.

But first half has $p + t$ a 's, second half has p a 's.

Since $p + t \neq p$, they cannot be reverses. Contradiction

Therefore, L is not regular.

Ques - 15c :

Prove $L = \{a^n \mid n \text{ is a perfect square}\}$ is not regular.

Proof:

Assume L is regular with pumping length p .

Consider $s = a^{p^2} \in L$.

By pumping lemma, $s = xyz$ where $|xy| \leq p$, $|y| = t > 0$.

Then $xy^2z = a^{p^2+t}$.

Next perfect square after p^2 is $(p+1)^2 = p^2 + 2p + 1$.

For $p^2 + t$ to be perfect square: $t \geq 2p + 1$.

But $t \leq p < 2p + 1$ for $p \geq 1$. Contradiction

Therefore, L is not regular.

Ques - 16 :

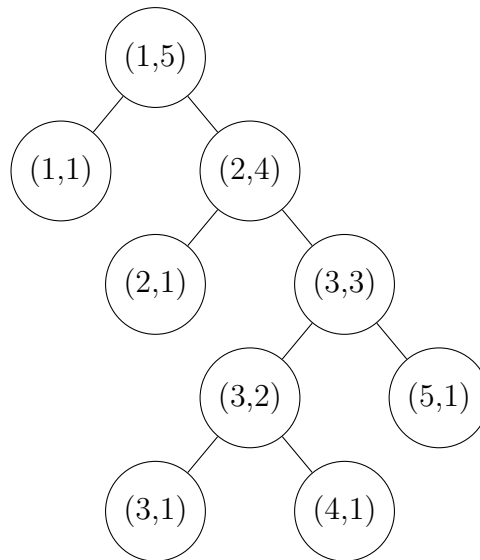
Parse the string using CYK algorithm shown in Table.

Solution:

Parsing Table:

$i \backslash j$	1	2	3	4	5
1	a	ϕ	k=1	k=2	k=1,2
2	b	k=1	k=2	k=1	NA
3	b	k=1	k=2	NA	NA
4	b	k=1	NA	NA	NA
5	b	NA	NA	NA	NA

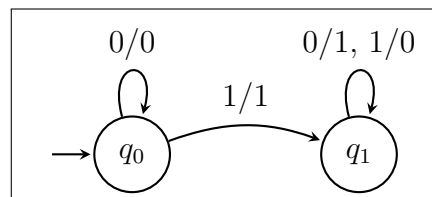
Parse Tree for "abbbb":



Ques- 17:

Design Mealy machine for 2's complement (LSB to MSB). Show both cases: carry discarded and not discarded.

Case 1: Carry Discarded



Case 2: Carry Not Discarded

